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TINY HOUSE AND A METHOD TO CONSTRUCT TINY HOUSE USING NARROW BUILDING PANELS

Abstract

A method to connect narrow building panels to provide a surface that has a visual resemblance of one solid panel is disclosed. Additionally, an assembly kit for building a tiny house of the narrow building panels with the connection method is disclosed as well as a sturdy tiny house having all surfaces made of the narrow building panels.

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Background/Summary

FIELD OF THE INVENTION

[0001] The invention is related to the method of building construction, especially to a system and method to connect building panels. Further, the invention relates to a system and method to connect

cross-laminated timber panels and to an assembly kit to build a house, specifically a tiny house.

BACKGROUND

[0002] Cross-laminated timber (CLT), also known as laminated timber, originates from Central Europe. The material began to be actively developed and produced in the mid-1990s in Austria and Germany, but its wider distribution and adoption have taken place over the last decade. CLT is made from vertically and horizontally laminated wood lamellas bonded together with adhesive, with the most common being 3-ply and 5-ply material. Spruce wood is commonly used as wood material.

[0003] As a market standard, large panels of Cross-Laminated Timber (CLT) are produced through the lamination of lamellas. The dimensions of these panels can reach up to 3 meters in height and 20 meters in width. The thickness generally varies from 60 mm to 300 mm, with exceptions reaching 500 mm. CLT's strengths compared to other commonly used construction materials include resistance to environmental impact, the ratio of weight to stiffness, environmental friendliness, and the speed of construction on the building site.

[0004] To achieve the last advantage, large CNC processing machines are used in production. These machines can mill entire walls of individual houses from large panels at once. This allows for the combination of strengths arising from controlled production environments, such as continuous production lines and high precision machines measured in millimeters. Ultimately, this reduces the most expensive and time-consuming component, which is the time spent on the construction site.

[0005] However, this production method results in higher material consumption, as all the cutouts from the panels essentially become production waste unless they can be repurposed into by-products. Various perforated steel brackets are generally used to connect the large panels. Without covering the panels with exterior insulation layers, these brackets will remain visible.

[0006] Using large panels for construction results in high transportation costs.

[0007] Tiny house movement is an architectural movement having a goal of downsizing and simplifying living spaces. In addition to simplified living spaces, eco friendliness is another aspect of the goals of the movement. Further goal of the movement is economically affordable construction of customized housing.

[0008] Factory-built houses have been transported as modules for relatively long, but this is not a very efficient solution as a significant portion of the shipment consists of air, and therefore large shipping containers are needed. On the other hand, if the house is built completely from start to finish at the construction site, the work requires skilled construction workers.

[0009] Accordingly, there is a need for tiny house production that would allow economic transportation, easy on-site assembly, and preferably use of cut out material from other construction industry. Moreover, the aesthetic aspects should be taken into account. Furthermore, as various kinds of natural disasters, including heavy storms, are especially destructive to tiny houses, there is a need for improved strength and solid structure for tiny houses as well as for other susceptible buildings, such as sheds, garages, and building extensions among others made of narrow building panels, e.g. CLT.

[0010] Consequently, there is a need for economical, durable, easy to build, eco-friendly methods and materials for building esthetically pleasing tiny houses.

SUMMARY

[0011] This disclosure provides improvements to the current technologies and at least some of the above mentioned issues. The methods, systems, and building assemblies disclosed and claimed here are specifically suitable to provide solutions for building tiny houses, but more generally the solutions disclosed in this application and claimed here are suitable for constructing any sizes of buildings.

[0012] Accordingly, the object of this disclosure is to provide a method to connect a multitude of narrow building panels, preferably CLT panels together to form a panel having a visual appearance

of being a solid panel, the method comprising the steps of: [0013] a) providing a first narrow building panel having a first and a second long side surfaces and a tongue and groove formation along a length of the second long side surface; [0014] b) providing a second narrow building panel having a mating tongue and groove formation along a first long side surface, and optionally having a second tongue and groove formation on a second long side surface opposite to the first long side surface, and at least one predrilled channel extending from an opening on the second side surface to a closed bottom inside the second narrow building panel; [0015] c) connecting the first and the second narrow building panels by connecting the tongue and groove formation of the first narrow building panel with the mating tongue and groove formation of the second building panel, whereby a connection zone having a middle section and two side sections is formed; [0016] d) inserting an attachment screw through the opening of the at least one predrilled channel and tightening the attachment screw through the bottom of the at least one predrilled channel such that a tip of the attachment screw penetrates a middle section of the connection zone into the first narrow building panel; [0017] e) optionally inserting further attachment screw through the openings of further predrilled channels and threading the further attachment screws through the bottoms of the further predrilled channels such that tips of the further attachment screws penetrate the middle section of the connection zone into the first narrow building panel; [0018] f) optionally providing a third narrow building panel having a mating tongue and groove formation on a first long side surface being configured to mate with the second tongue and groove formation of the second narrow building panel, and at least one predrilled channel extending from an opening on the second long side surface to a closed end inside the third narrow building panel and being located in a staggered manner in comparison to location of the predrilled channels of the second narrow building panel; [0019] g) connecting the second and the third narrow building panels by connecting the second tongue and groove formation of the second narrow building panel with the mating tongue and groove formation of the third narrow building panel, whereby a connection zone having a middle section and two side sections is formed; [0020] h) inserting an attachment screw through the opening of the at least one predrilled channel of the third narrow building panel and tightening the attachment screw through the bottom of the at least one predrilled channel such that a tip of the attachment screw penetrates the middle section of the connection zone in a staggered manner above or below a level of the predrilled channel(s) of the second narrow building panels; [0021] i) optionally inserting further attachment screws through the openings of further predrilled channels and threading the further attachment screws through the bottoms of the further predrilled channels such that tips of the further attachment screws penetrate the middle section of the connection zone into the second narrow building panel; and [0022] j) optionally repeating steps f) to i) until all narrow building panels of the multitude of narrow building panels are attached together.

[0023] In the method the tongue and groove formation preferably comprises two grooves and one tongue along a long side surface of one panel and two tongues and one groove along a long side surface of another panel to be connected.

[0024] The method may comprise a step of securing the tongue and groove formations with an adhesive.

[0025] At least part of the narrow building panels may additionally be provided with a short positioning channel predrilled on a long surface side of the panels opposite to each predrilled channel to guide tightening of the attachment screws.

[0026] The narrow building panels may have a width of 350 to 600 mm, preferably 400 to 560 mm, and most preferably 410 to 500 mm.

[0027] In certain aspects of the method, the narrow building panel is a narrow CLT panel, and the predrilled channels are drilled in a direction perpendicular to the long side surface through which the channel is drilled, and the predrilled channels are completely within a middle layer of the narrow CLT panels.

[0028] In certain aspect of the method the predrilled channels are drilled in an angle deviating from

a straight angle through the long side surface. The angle may be 91-140 degrees, preferably 100-125 degrees, most preferably 105-115 degrees.

[0029] The method of this invention may be for making surface panels for a tiny house.

[0030] It is a further object of this invention to provide a surface panel for construction of a building, wherein the panel is according to the method described here. The surface panel may be a surface panel for a tiny house. The surface panel may be made by connecting narrow CLT panels together according to the method of this disclosure, but the material may also be wood, laminated wood products (glulam), concrete, plastic, composite material, plywood or other material acceptable as bearing construction material.

[0031] Yet another object of this invention is to provide a method to construct a tiny house, wherein the method comprises providing a floor surface, multiple wall surfaces, and a roof surface of narrow building panels attached together according to the attachment method disclosed here.

[0032] An object of this invention is to provide an assembly kit for construction a tiny house, wherein the kit comprises: [0033] a predetermined number of first narrow building panels having a groove and tongue formation on one long side surface of the panels; [0034] a predetermined number of second narrow building panels having a groove and tongue formation on both long side surfaces of the second panels and one or more predrilled channels on one of the long side surfaces, [0035] wherein the predrilled channels of at least two of the second narrow building panels are on different levels and the second narrow building panels are numbered to indicate an order of assembly of the second panels such that two second panels having the channels on same level are never attached next to each other.

[0036] The assembly kit provided here may further comprise window and door structures, and optionally collapsible stair structures along with assembly instructions.

[0037] An object of this invention is further to provide a tiny house having surface structures made of narrow building panels, preferably narrow CLT panels, connected together with the method disclosed herein. The surface structures may be a floor structure, four wall structures, interior wall structures, a roof structure and a floor structure of a loft.

[0038] An object of this invention is to provide a method to construct a tiny house from narrow building panels, preferably narrow CLT panels, and the tiny house constructed of the narrow building panels according to the method without using any visible brackets, wherein the tiny house has increased flexural strength and shear force of the connection.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0039] FIG. **1a** illustrates attaching two narrow building panels (**10**) F-1 and F-2 to each other along their long sides having connecting groove and tongue formations (**20**). F-1 is the first panel and is attached only from one of its long sides to the second panel (F-2). For this reason, F-1 has the groove and tongue formation (**20**) only on the side where F-2 is to be attached. The groove and tongue formation of the second panel F-2 is configured to be mating with the formation of the first panel F-1 on the long side. The figure also shows attaching the first panel F-1 onto the floor studs (**50**) with screws (**52**).

[0040] FIG. **1b**. show a cross sectional view from the direction of the short ends of the narrow building panels F-1 and F-2. Here, the flat long side (**54**) of the first panel F-1 which does not have a tongue and groove formation is visible. F-1 and F-2 are connected along their long side surfaces with the groove and tongue formation (**20**) where the tongues (**20a**) of F-1 match with grooves (**20b**) of F-2. In the shown embodiment, one panel has two grooves and a tongue in between while the mating panel has two tongues and a groove in between. The number of grooves and tongues may be different though as long as the tongue and groove formations of the panels to be attached

together are mating with each other along the long side surfaces. Connecting the tongue and groove formations forms a connection zone (20c, shown in FIG. 2b) having a middle section (20d, shown in FIG. 2b). A cross sectional cut of a predrilled channel (40) is shown on F-2. The channel is drilled on the long side surface of second panel F-2 such that the opening of the channel (21) is located in the groove of the groove and tongue formation. The bottom of the channel (42) is inside the second panel F-2. An attachment screw (30) is inserted through the predrilled channel and tightened so that the head of the attachment screw is at the bottom of the predrilled channel and the threaded tip (34) of the attachment screw has penetrated the middle section of the groove and tongue formation and extends into the first panel F-1 thereby providing stable and strong attachment to the panels without use of any brackets.

[0041] FIG. 2 shows attachment of the narrow building panels with multiple concealed attachment screws (30). a) is a front view showing two attachment screws being inserted through a predrilled channel (40) of the left hand side panel and tightened such that the head (32) of the attachment screw is at the bottom of the predrilled channel. The tip of the attachment screw extends out from the opposite side of the panel through to the right hand side panel. The width of the panel is indicated with element number 12. b) is a perspective view showing the attachment. Here the connecting tongue and groove formations along the long sides of the panels are shown to form the connection zone (20c). The middle section of the connection zone is indicated with 20d. The length of the panel is indicated with element number 14.

[0042] FIG. 3 illustrates the staggered manner of location of the predrilled channels and attachment screws (30) in attachment of the narrow building panels. The predrilled channels (40) of two adjacent panels are on different levels. In a) an attachment screw has been inserted through a predrilled channel of the right hand side panel and tightened such that the head of the attachment screw is in the bottom of the predrilled channel and the tip of the attachment screw penetrates through the panel into next panel (not shown here). On the left hand side panel of figure a) the panel has a predrilled channel made on its long side surface and an attachment screw is inserted into the channel but not tightened yet. In b) on the left hand side panel the second attachment screw is tightened through the channel such that the head of the attachment screw is in the bottom of the channel and the tip of the attachment screw penetrates though the panel and extends out from the opposite long side through to the adjacent panel on the right hand side and thus attaches the panels together. Importantly the predrilled channel is drilled such that when the panels are attached to each other the predrilled channels of the adjacent panels are never at same level; i.e., they are staggered. The figures illustrate how the attachment screws locate within the adjacent panels in a staggered manner. The figures also show short positioning channels (44) drilled on the second long side of the panels directly opposite to predrilled channels (40). The arrows in the figures illustrate the direction of building a larger panel by attaching the narrow building panels together.

[0043] FIG. 4 is an illustration of installment of wall panels. The installation starts in the corner with panels WB7 and W26, these first panels are attached to each other with screws (not shown) similarly as F-1 in FIG. 1a is attached to a stud. Element W25 has a predrilled channel through which an attachment screw is installed similarly as shown in FIG. 2, and the tip of the attachment screw penetrates through element W26 thereby attaching elements W26 and W25. Element W24 has predrilled channel through which attachment screw is installed to penetrate element W25 thereby attaching element W24 and W25 together and so on. Same logic applies to elements WB6, WB5 etc. The arrows implicate the order of attaching the panels to each other. In an assembly kit for a tiny house, the narrow building panels are numbered, and instructions are provided to the order of installment.

[0044] FIG. 5 is an illustration showing how a window panel is inserted in place. Once the window panel is inserted on top of the short wall panels (WB5-WB6), longer panel WB4 is attached by inserting attachment screws through the predrilled channels (openings of two channels shown here on the long side surface of panel WB4) and the tip of the attachment screws penetrate to the short

panel WB5 thereby attaching the long panel WB4 to the short panel WB5.

[0045] FIG. 6 is an illustration where short panels W14 and W14-1 are attached on top of a window. Here two openings of predrilled channels are shown on the long sides of these short panels. Attachment screws are inserted through the predrilled channels of element W14 so that the attachment screw tips penetrate the long panel next to the window and thus attaches W14 to the wall structure already assembled. The predrilled channels of element W14-1 are positioned in a staggered manner so that they are not at same level as the predrilled channels of panel W14. The attachment screws are installed through the predrilled channels of W14-1 so that their tips penetrate through W14, thereby attaching W14-1 and W14. The long panel W16 is shown to have four predrilled channels that are drilled of a long side surface of W16 in a staggered manner such that they do not coincide with the channels of W14-1 or W15-1, and W16 is attached by attachment screws installed into the channels. This figure shows how the narrow building panels may be installed horizontally, or diagonally (above the window): the manner of connecting the narrow building panels is still along their long sides.

[0046] FIG. 7 illustrates attachment of roof channels of a tiny house. The first panel R1 is attached by screws on top of the installed wall panels, similarly as the first floor panel to the studs in FIG. 1a. The rest of the panels have predrilled channels in a staggered manner such that attachment screws installed through the channels will not coincide with position of the predrilled channels in the adjacent previous panel. In the figure four openings of the predrilled channels are shown in each panel.

[0047] FIG. 8 is a perspective view of a tiny house where the roof, walls and floor are all assembled by using narrow building panels with attachment method described here without use of any brackets.

[0048] FIG. 9 illustrates attachment of narrow building panels with multiple concealed attachment screws in an angled manner. a) is a front view showing an attachment screw (30) inserted through a predrilled channel (40) drilled in an angle (a). In this specific case the predrilled channel extends till about two-thirds of the width the left hand side panel in the figure and is tightened such that the head of the attachment screw is at the bottom of the channel. The tip of the attachment screw extends out from the opposite side of the panel attaching the two panels together. b) shows the two panels attached together with the groove and tongue formation and with two attachment screws inserted through the predrilled channels that are drilled in an angle (a) till about $\frac{2}{3}$ of the width of the panel (left hand side panel in the figure) and tightened such that the head of the attachment screw is at the bottom of the channel. The tips of the attachment screws penetrate to the adjacent second panel (on the right hand side in the figure) through a middle section (20d) of a connection zone (20c) formed by the mating groove-tongue structures. The figure shows also side sections (20c) of the connection zone. c) is a cross sectional view showing the attachment screw having been inserted into the predrilled channel drilled in an angle and tightened so that its head is at the bottom of this channel and the tip of the attachment screw penetrates to the right hand side panel through the middle section (20d) of the connection zone formed by matching groove-tongue formation. d) is a front view of the groove-tongue structure connecting two adjacent panels. The figure illustrates the connection zone having a middle section and two side sections.

[0049] FIG. 10 shows four narrow building panels attached together with the groove and tongue formations and with two attachment screws inserted through predrilled channels that are located in staggered manner: the predrilled channels in every second panel are on same level this illustration. a) shows an example where the channels are drilled perpendicularly against the long side surfaces of the panels, while b) shows an example where the channels are drilled in an angle different from a straight angle. Both in a) and in b) the channels are drilled till about $\frac{2}{3}$ way of the width of the panel and tightened such that the head of the attachment screw is at the bottom of the channel. The channels and the attachment screws within the channels are completely inside the panels and therefore the inner and outer surface of the panel structure created has a visual impression of being

a uniform single panel. While the figures a) and b) show panels where the narrow building panels are connected together and are in vertical position, similar panels can be achieved by connecting the narrow building panels together in horizontal or even diagonal position; the long side surfaces of the narrow building panels are always connected together through predrilled channels located in staggered manner.

DETAILED DESCRIPTION

[0050] This disclosure describes a method to assemble a tiny house using narrow building panels by connecting them by concealed attachment screws without use of any brackets. Disclosed also is an assembly kit for building a tiny house. Furthermore, a system to connect narrow building panels together such that the surfaces of the connected panels have a visual resemblance of one large panel is disclosed here. The connection system can be used to form surface panels for tiny houses but also for any other type of buildings. Notably the method, system and assembly kit disclosed and claimed here do not use any kind of visible brackets.

[0051] The term “narrow building panel” as used here means panels made of material for example wood or laminated wood products (glulam), concrete, plastic, composite materials, plywood or other material allowed as bearing construction material and having a narrow width as defined below. Most preferably the material of the narrow building panels is CLT. The narrow building panels, regardless of the material they are made of are 200-800 mm wide. The width (**12** in FIG. 2a) of the “narrow building panel” may be 200-600 mm, it may be 300-600 mm, it may be 400-600 mm, 400-500 mm, and in certain embodiments the narrow building panel may have a width of 420 mm to 440 mm, for example 430 mm. The length (**14** in FIG. 2 b) of the narrow building panels may vary from 200 to 8000 mm, but other lengths may also be used. In case the narrow building panel is a narrow CLT panel, the narrow CLT panels have 3 or 5 ply lamellas, most preferably the panels have 3 ply lamellas. The thickness of the panels is preferably 40 to 300 mm.

[0052] The term “attachment screw” as used here, means a threaded or non-threaded attachment element. A threaded attachment element is an actual screw to be tightened by a screwdriver, while a non-threaded attachment element is a nail that can be set in place by applying force with a hammer on a metal rod placed on the head of the element. Using a non-threaded nail in the attachment may be beneficial alternative in situations where electricity is not available for powered screwdriver, for example in forest environments.

[0053] The term “tiny house” is generally understood to mean a dwelling that has maximally an area of about 50 square meters. However, even if this disclosure uses the term ‘tiny house’ the term is not meant to limit the type or size of buildings constructed fully or partially with the methods and systems of this application and claims, but the methods, systems and assemblies provided in this application can be used to connect narrow building panels together for any size of buildings.

[0054] Moreover, the building constructed with the methods and systems of this application does not necessarily need to be a dwelling, but may as well be a shed, a garage or other non-dwelling structure. Furthermore, the building may as well be an extension of an existing building. The methods and systems provided here may be used to connect at least part of the narrow building panels of a surface of a building that has an area up to 2000 square meters.

[0055] The method described here in detail with reference to the figures allows building a tiny house with an aesthetic look with a uniform surface giving an impression of each surface of the tiny house being built of one large panel. Moreover, the method described here enables providing all the material and instructions needed for construction of a tiny house in a flat tightly packed package, i.e., avoiding empty space in the package. Assembly of the tiny house using the assembly kit described and claimed here can be completed in its entirety by two persons without any specific construction skills and with simple tools such as a hammer, and a powered screwdriver.

[0056] The method to assemble the tiny house as described here is based on the manner in which the narrow building panels, preferably narrow CLT panels, are attached to each other.

[0057] As shown in FIG. 1a, a starting point to assemble any surface of the tiny house is having

two narrow building panels (**10**) each having two long side surfaces. Each one of the narrow building panels has at least one of the two long sides milled to have a groove and tongue formation (**20**) along the whole long side surface and the groove and tongue formations are configured to match each other. As shown in FIG. **1b**, the groove and tongue formation of one panel may have two tongues and a groove in between and the matching groove and tongue formation of the other panel has two grooves and a tongue in between. It is possible to have any other number of grooves and tongues as long as grooves of one panel match the tongues of the other one. The preferred embodiment is as shown in FIGS. **1b** and **1n** FIG. **9d** for a panel where the narrow building panels are formed of three plies (lamellas). In this case, each of the grooves and each of the tongues are configured to be located completely within one ply. The first step in connecting the two narrow building panels is preferably to connect the matching groove and tongue formations with an adhesive. Connection of the matching groove and tongue formations form a connection zone (**20c**) shown e.g. in FIGS. **9b** and **d**). The connection zone has a middle section (**20d**) and two side sections (**20c**). The middle section of the connection zone is in the middle lamella of the narrow building panel.

[0058] One of the two narrow building panels is provided with one or more predrilled channels (**40**) on the long side surface opposite to the surface having the groove and tongue formation that is connected to adjacent panel as described above preferably with an adhesive. The channels may be drilled perpendicularly against the long side surface (FIG. **3**). Alternatively, the channels may be drilled diagonally against the long side surface (FIG. **9 a, b**). In this case the angle (α , shown in FIG. **9a**) between the long side surface and the channel is larger than 90 degrees but smaller than 180 degrees. In certain aspects the angle may be larger than 90 degrees but smaller than 150 degrees. In certain aspects the angle may be in the range of 91 to 140 degrees. In certain aspects the angle may be in a range of 100-125 degrees, and in certain aspects between 105 and 115 degrees.

[0059] The channel is drilled in a manner that it extends inside the middle section (**20d**) through to about $\frac{1}{4}$ to $\frac{3}{4}$ of the width of the narrow building panel; more preferably about $\frac{1}{2}$ - $\frac{2}{3}$ width of the narrow building panel. The depth of the channel depends on the length of the attachment screw (**30**). The length of the attachment screw, which as defined here may be a threaded screw or a non-threaded nail, may be 80 to 500 mm, preferably 150-300 mm, and most preferably 250-300 mm, for example such as 260 mm. In one preferred embodiment the attachment screw is 260 mm long and the depth of the channel is 280 mm; however, the depth of the channel may be 150-650 mm.

Notably the same length of attachment screws may be used for panels of different widths: for a wider panel the predrilled channel would be deeper, but the attachment screw length may be same as with narrower panels. The channel (**40**) is configured to receive the attachment screw (**30**). The attachment screw is tightened or otherwise set inside the channel through the bottom of the channel through the middle section (**20d**) of the connection zone into the adjacent narrow building panel.

The length of the attachment screw is preferably such that the tip of the attachment screw penetrates about $\frac{1}{4}$ - $\frac{3}{4}$, and preferably $\frac{1}{4}$ - $\frac{1}{2}$ of the width of the adjacent narrow building panel. Most preferably the attachment screw penetrates inside the adjacent narrow building panel to about one third of the width of the adjacent narrow building panel. The predrilled channel has a diameter that is determined by the head size of the attachment screw. In a preferred embodiment, the narrow building panels is a narrow CLT panel, and the head size of the screw is the same as the width of the middle lamella of the CLT panel. As is shown in FIG. **3 a,b**, in addition to the predrilled channels (**40**) adapting the attachment screw, there may be predrilled short positioning channels (**44**) on the opposite long side surface of the panel. The purpose of these positioning channels is to ensure that the attachment screw when tightened through the predrilled channel stays within the middle section (**20d**). It is possible to attach the two narrow building panels with only one attachment screw, but more preferably the two panels are attached with multiple attachment screws. In other words, one of the narrow building panels have multiple channels predrilled through the long side surface. Preferably the distance between each predrilled channel is 100-3000 mm,

preferably 600-1800 mm and most preferably 1100-1300 mm, such as 1200 mm. Most preferably attachment of two adjacent narrow building panels is made by at least 2 attachment screws, such as 2 to 12, 3 to 10, or 6 to 10 attachment screws depending on the length of the narrow building t one of the two narrow building panels that are described above to be connected via the groove and tongue formation and the attachment screws concealed in the predrilled channels has a groove and tongue formation on the second long side surface also (FIG. 1b). This surface is configured to be connected to a third narrow building panel which has a mating groove and tongue formation on one of its long side surfaces (see FIG. 3). The third narrow building panel is first attached to the previous narrow building panel by matching the tongue and groove formation preferably with an adhesive. After this the third narrow building panel is attached to the previous narrow building panel with concealed attachment screws through predrilled channels in the second long side surface of the third narrow building panel. Importantly, now the predrilled channels of the third narrow building panel are located such that when the third panel is attached to the previous panel the channels are on different levels. This way the screws when inserted through the channels to penetrate the previous narrow building panel will never be on same level; in other words, the screws are staggered as is shown in FIGS. 3 and 10.

[0060] The third panel may again have another groove and tongue formation along the second long side surface and a fourth panel may be attached to the third one similarly as the third was attached to the second one. Again, the predrilled channels in the fourth narrow building panel are located such that when the fourth panel is attached to the previous one the channels are on different levels (staggered). The channels on the fourth narrow building panel may be on same level as the channels of the second narrow building panel, but the channels may be drilled in a different level than any of the channels of the previous panels. FIG. 10 shows an example where every second panel has the predrilled channels on the same level. In one referred embodiment every 3rd panel has the channels drilled on the same level.

[0061] When assembling the tiny house, the first narrow building panel of each surface (i.e. roof surface, wall surface, floor surface) is attached to studs with regular screws or nails (see FIG. 1a). The first narrow building panel of each surface therefore only has one long side surface having the groove and tongue formation, the other long side of the first panel is a flat surface (54 in FIG. 1b).

[0062] The method to connect the narrow building panels to each other as described above enables a full assembly kit for construction of a tiny house. The kit comprises a predetermined number of first narrow building panels having a groove and tongue formation on one long side surface of the panels and a predetermined number of second narrow building panels having a groove and tongue formation on both long side surfaces of the panels. The second narrow building panels having a groove and tongue formation on both long side surfaces also have one or more predrilled channels on one of the long side surfaces. The predrilled channels of at least two of the narrow building panels are on different levels and the narrow building panels are numbered to indicate an order of the panels such that two panels having the channels on same level are never attached next to each other. Preferably the assembly kit comprises for each surface to be built, one first narrow building panel and a multitude of second narrow panels configured to be connected to each other to form the surface panel where the narrow second building CLT panels have the predrilled channels on at least two different levels such that the channels of two second narrow building panels to be connected together are never on same levels. In addition to the narrow building panels, the kit comprises window and door structures as well as optional collapsible stair structures along with assembly instructions.

[0063] Due to the narrow width of the building panels in this disclosure, the assembly kit can be fitted into a single flat package avoiding any empty space in the package. The package may have dimensions of 430-2300 mm by 1000-12000 mm by 130-2300 mm. A preferable size is 1000 mm×1000 mm×4000 mm; however, it is to be understood that the size of the package depends on the design of the house to be built. Another advantage of the narrow width of the building panels of

this disclosure is that at least part of the panels in the assembly kit may be made by cut off material of other industry. Preferably, the narrow building panels are narrow CLT-panels made of cut off material.

[0064] Assembly of the tiny house from the assembly kit is designed such that two average skilled persons without any special tools can assemble the whole house. The assembly of the house in good conditions may take just one full working day.

[0065] The house built from the assembly kit according to the instructions is sturdy by its structure. Having the predrilled channels for the attachment screws drilled in an angle (a) which is different from a straight angle in relation to the long side surface in at least part of the narrow building panels is greatly improving the stability of the tiny house. In case of narrow CLT-panels changing the angle of the pre-drilled channel to deviate from 90 degrees angle allows the attachment screw to penetrate the middle lamella in a direction perpendicular to grain of the lamella. The angle of the channel has an effect on flexural strength and shear force of the connection and increased values indicate higher stability of the tiny house structure in high winds.

ELEMENT NUMBERS

[0066] **10** narrow building panel [0067] **10a** a first narrow building panel [0068] **10b** a second narrow building panel [0069] **12** width of the narrow building panel [0070] **14** length of the narrow building panel [0071] **20** connecting groove and tongue formation of the narrow building panels [0072] **20a** tongue [0073] **20b** groove [0074] **20c** connection zone [0075] **20d** middle section of the connection zone [0076] **20e** side section of the connection zone [0077] **21** opening of the predrilled channel [0078] **30** attachment screw [0079] **32** head of the attachment screw [0080] **34** threaded tip of the attachment screw [0081] **40** predrilled channel for the attachment screw [0082] **42** bottom of the predrilled channel [0083] **44** short positioning channel [0084] **50** floor stud [0085] **52** screw [0086] **54** flat side surfaces of the first floor panel

Claims

1. A method to connect a multitude of narrow building panels together to form a panel having a visual appearance of being a solid panel, the method comprising the steps of: a) providing a first narrow building panel having a first and a second long side surfaces and a tongue and groove formation along a length of the second long side surface; b) providing a second narrow building panel having a mating tongue and groove formation along a first long side surface, and optionally having a second tongue and groove formation on a second long side surface opposite to the first long side surface, and at least one predrilled channel extending from an opening on the second side surface to a closed bottom inside the second narrow building panel; c) connecting the first and the second narrow building panels by connecting the tongue and groove formation of the first narrow building panel with the mating tongue and groove formation of the second building panel, whereby a connection zone having a middle section and two side sections is formed; d) inserting an attachment screw through the opening of the at least one predrilled channel and tightening the attachment screw through the bottom of the at least one predrilled channel such that a tip of the attachment screw penetrates a middle section of the connection zone into the first narrow building panel; e) optionally inserting further attachment screws through the openings of further predrilled channels and threading the further attachment screws through the bottoms of the further predrilled channels such that tips of the further attachment screws penetrate the middle section of the connection zone into the first narrow building panel; f) optionally providing a third narrow building panel having a mating tongue and groove formation on a first long side surface being configured to mate with the second tongue and groove formation of the second narrow building panel, and at least one predrilled channel extending from an opening on the second long side surface to a closed end inside the third narrow building panel and being located in a staggered manner in comparison to location of the predrilled channels of the second building panel; g) connecting the second and the

third narrow building panels by connecting the second tongue and groove formation of the second narrow building panel with the mating tongue and groove formation of the third building panel, whereby a connection zone having a middle section and two side sections is formed; h) inserting an attachment screw through the opening of the at least one predrilled channel of the third narrow building panel and tightening the attachment screw through the bottom of the at least one predrilled channel such that a tip of the attachment screw penetrates the middle section of the connection zone in a staggered manner above or below a level of the predrilled channel(s) of the second building panels; i) optionally inserting further attachment screws through the openings of further predrilled channels and threading the further attachment screws through the bottoms of the further predrilled channels such that tips of the further attachment screws penetrate the middle section of the connection zone into the second narrow building panel; and j) optionally repeating steps f) to i) until all narrow building panels of the multitude building panels are attached together.

2. The method of claim 1, wherein the tongue and groove formation comprises two grooves and one tongue along a long side surface of one panel and two tongues and one groove along a long side surface of another panel to be connected.

3. The method of claim 1, wherein the tongue and groove formations are secured by an adhesive.

4. The method of claim 1, wherein at least part of the narrow building panels are additionally provided with a short positioning channel predrilled on a long surface side of the panels opposite to each predrilled channel to guide tightening of the attachment screws.

5. The method of claim 1, wherein the narrow building panels have a width of 350 to 600 mm, preferably 400 to 560 mm, and most preferably 410 to 500 mm.

6. The method of claim 1, wherein the narrow building panels are narrow CLT panels.

7. The method of claim 6, wherein the predrilled channels are drilled in a direction perpendicular to the long side surface through which the channel is drilled, and the predrilled channels are completely within a middle layer of the narrow CLT panels.

8. The method of claim 1, wherein the predrilled channels are drilled in an angle deviating from a straight angle through the long side surface.

9. The method of claim 8, wherein the narrow building panels are narrow CLT panels and the predrilled channels are completely within a middle layer of the narrow CLT panels.

10. The method of claim 8, wherein the angle is 91-140 degrees, preferably 100-125 degrees, most preferably 105-115 degrees.

11. The method of claim 1, wherein the method is for making surface panels for a tiny house.

12. A surface panel for a construction of a building, wherein the panel is made according to claim 1.

13. The surface panel of claim 12, wherein the surface panel is a surface panel for a tiny house.

14. A method to construct a tiny house, the method comprises providing a floor surface, multiple wall surfaces, and a roof surface of narrow building panels attached together according to the method of claim 1.

15. An assembly kit for construction a tiny house, the kit comprising: a predetermined number of first narrow building panels having a groove and tongue formation on one long side surface of the panels; a predetermined number of second narrow building panels having a groove and tongue formation on both long side surfaces of the second panels and one or more predrilled channels on one of the long side surfaces, wherein the predrilled channels of at least two of the second narrow building panels are on different levels and the second narrow building panels are numbered to indicate an order of assembly of the second panels such that two second panels having the channels on same level are never attached next to each other.

16. The assembly kit of claim 15, wherein the narrow building panels are narrow CLT panels.

17. The assembly kit of claim 15, wherein the kit further comprises window and door structures, and optionally collapsible stair structures along with assembly instructions.

18. A tiny house having surface structures made of narrow building panels, preferably of narrow

CLT panels, connected according to claim 1.

19. The tiny house of claim 18, wherein the surface structures are a floor structure, four wall structures, interior wall structures a roof structure and a floor structure of a loft.

20. A tiny house having surface structures made of narrow CLT panels connected according to the method of claim 7, wherein the tiny house has increased flexural strength and shear force of the connection.
